

Tutorial No.6

Q.1) Apply the Gram-Schmidt orthogonalization process to find orthogonal basis, for the set of the vectors

$$\text{i) } S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \right\}$$

$$\text{ii) } S = \left\{ \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 8 \end{bmatrix} \right\}$$

$$\text{iii) } S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Q.2) Apply the Gram-Schmidt orthonormalization process to find orthonormal basis, for the set of the vectors

$$\text{i) } S = \{v_1 = t^2, v_2 = t^3\} \text{ Where } \langle u, v \rangle = \int_0^1 uv dt$$

$$\text{ii) } S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 8 \end{bmatrix} \right\}$$

Q.3) Find the least square solutions of $AX = B$ where:

$$A = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

